1 Introduction to contract testing

This chapter covers

- What this book is about and why it’s important
- A high-level overview of contract testing
- Why you should care about contract testing
- When contract testing is appropriate to use and not use
- How contract testing works on a high level
- Contract testing in action

Imagine this scenario. It’s 4 p.m. on a Friday, and you decided to deploy a small change of updating one of your API’s data responses. All the automated tests from your pipeline are returning green, which gives you the confidence that you haven’t broken anything. You hit deploy, and a few minutes after the deployment, you get a message from another team that something is broken in production. It couldn’t be the change you recently deployed because it was a small change, and all the tests are passin… . right? As it turns out, this other team using your API has certain expectations. During the incident, you’ve learned that these expectations are different from yours. Rather than enjoying your Friday afternoon, you’re left with a production bug you must deal with.

During the incident postmortem, you identify that there are API integration tests written by this other team that would have caught this problem. Still, your change didn’t trigger those tests because it’s in a separate project repository and pipeline. On top of that, these API integration tests are notorious for being flaky and slow. One of the action items from the postmortem is to find alternatives to better catch these changes, and one of the suggestions was to look into contract testing. You’ve heard this term before, but you’re not sure how this will help with your problem.

Well, you’ve come to the right resource! This book covers all the important things you need to know about contract testing, such as what it is, how it